

Chemistry Basics for Reading Lunar Rocks

The Destination

This booklet aims at one target: understanding the four minerals that make up the Moon. By the end, this table will be completely readable to you:

Mineral	Formula	Where on the Moon	Why it matters
Plagioclase	$\text{CaAl}_2\text{Si}_2\text{O}_8$	Highlands	Magma-ocean hypothesis
Pyroxene	$(\text{Ca, Mg, Fe})\text{SiO}_3$	Maria	Backbone of lunar basalt
Olivine	$(\text{Mg, Fe})_2\text{SiO}_4$	Maria and mantle	Window into the deep Moon
Ilmenite	FeTiO_3	Ti-rich maria	Lunar base

Vocabulary:

A **mineral** is a single crystalline compound with a definite chemical formula.

A **rock** is an aggregate of mineral grains stuck together.

Anorthosite is a *rock* made of >90% plagioclase grains.

Basalt is a *rock* made mostly of pyroxene + plagioclase ($\pm$ olivine, $\pm$ ilmenite). Quick test: if it has a formula, it's a mineral.

Contents

The Destination	1
1 The Toolkit	5
1.1 What chemistry is	5
1.2 Numbers you'll need	5
1.3 States of matter	5
2 Atoms and the Eight Elements	6
2.1 The anatomy of an atom	6
2.2 What makes an element an element	6
2.3 Atomic mass	6
2.4 The periodic table is a map, not a list	6
2.5 The Lunar Eight	6
2.6 The Moon is an oxygen warehouse	7
3 Electrons, Ions, and the Logic of Charge	8
3.1 Electron shells: seating around the nucleus	8
3.2 The octet rule: atoms want full shells	8
3.3 Ion size: the second rule of minerals	8
4 Bonds, Formulas, and Charge Balance	10
4.1 Two kinds of glue	10
4.2 How to read a formula	10
4.3 The charge-balance	10
4.4 Formula mass	11
5 The Silicate Tetrahedron	12
5.1 Why silicon and oxygen rule the crust	12
5.2 Meet the tetrahedron	12

5.3	Corner-sharing: the master variable	12
5.4	What each architecture implies	12
5.5	The odd one out: ilmenite the oxide	13
6	Crystals: Architecture Determines Everything	14
6.1	What a crystal actually is	14
6.2	The parking-map view of minerals	14
6.3	Cleavage: crystals break where bonds are weak	14
6.4	Density: the property that will shape the Moon	14
6.5	Color: why the Moon has a face	15
7	Moles and Stoichiometry: Counting Atoms by Weighing Rocks	16
7.1	The counting problem	16
7.2	The mole	16
7.3	Subscripts are mole ratios	16
7.4	Balancing equations	16
7.5	Reading a real lunar analysis: oxide weight-percent	17
8	Solid Solutions: What the Parentheses Mean	18
8.1	Minerals are families, not fixed compounds	18
8.2	Simple substitution: olivine's Mg ↔ Fe slider	18
8.3	Coupled substitution: plagioclase's elegant trick	18
8.4	The payoff: one number that records the Moon's birth	19
9	Redox: The Moon's Missing Rust	20
9.1	Electron bookkeeping	20
9.2	Iron's three costumes	20
9.3	The audit meets redox	20
9.4	Redox in reactions: reading the hand-off	21
10	The Magma Ocean: Why the Moon Has a Face	22
10.1	The setup: a molten Moon	22
10.2	Freezing is sequential, and each mineral edits the melt	22

10.3 Density is destiny	22
10.4 Act two: the maria (the part everyone gets backwards)	22
10.5 The grand synthesis	23
A The Booklet on One Page	24
B Master Data Tables	25

1 The Toolkit

1.1 What chemistry is

Chemistry is the study of **matter**, anything that has mass and takes up space, and how it transforms. A lunar rock is matter; and so is the oxygen we breathe. Chemistry's central claim, which took humanity 2,500 years to prove, is this: **All matter is made of a small number of ingredient types (elements), combined in countable whole-number ratios.** When you see $\text{CaAl}_2\text{Si}_2\text{O}_8$, you are seeing: 1 calcium, 2 aluminums, 2 silicons, 8 oxygens. Chemistry is arithmetic applied to atoms.

1.2 Numbers you'll need

Scientific notation. Atoms are absurdly small and absurdly numerous, so we write big numbers as powers of ten. 6.022×10^{23} means 6.022 with the decimal moved 23 places right. 1×10^{-10} means 0.0000000001—roughly the width of one atom in meters.

Units. We use grams (g), kilograms (1 kg = 1000 g), and grams per mole (g/mol). 1 mole = 6.022×10^{23} particles (Avogadro's number), and one mole of anything weighs its formula mass in grams: 1 mol O = 16.0 g; 1 mol Fe = 55.8 g; 1 mol ilmenite = 151.7 g. Always write units next to numbers; half the errors in chemistry are unit errors.

Percentages. "42% by weight" means: out of every 100 grams, 42 grams. To find 42% of 250 g: $0.42 \times 250 = 105$ g.

Ratios. If a recipe needs 2 aluminum for every 1 calcium, then 10 calciums need 20 aluminums. Mineral formulas are recipes; you will scale them constantly.

Exercises:

1. Write 15,000,000 in scientific notation.
2. What is 31.6% of 500 g?
3. A formula needs 3 oxygens per titanium; how many oxygens for 12 titaniums?

Answers: (a) 1.5×10^7 (b) 158 g (c) 36.

1.3 States of matter

Matter comes as solid, liquid, gas. The Moon has essentially no atmosphere and no liquid water; lunar chemistry is overwhelmingly **solid-state chemistry**, plus one crucial liquid: **magma** (molten rock).

2 Atoms and the Eight Elements

2.1 The anatomy of an atom

Every atom has:

1. **The nucleus:** a tiny, dense core containing **protons** (charge $+1$, mass ≈ 1 unit) and **neutrons** (no charge, mass ≈ 1 unit).
2. **The electron cloud:** a vast, mostly-empty region where **electrons** (charge -1 , mass ≈ 0.0005 units, effectively nothing) surround the nucleus.

Scale matters: if the nucleus were a marble at the center of a football stadium, the electrons would be gnats in the top rows. An atom is almost entirely empty space; yet you can't walk through a wall, because electron clouds refuse to overlap. A neutral atom has **equal protons and electrons**, so charges cancel.

2.2 What makes an element an element

The **number of protons**, the **atomic number**, Z , is an atom's identity card. It cannot change without nuclear physics (which minerals don't do). Every atom with 8 protons *is* oxygen; every atom with 26 *is* iron. Change the proton count: different element. Change the neutron count: same element, slightly heavier. Change the electron count: an **ion**, same element, now electrically charged.

2.3 Atomic mass

The **atomic mass** (in g/mol $\approx$) is roughly protons + neutrons. Oxygen: $8 + 8 \approx 16.0$. Iron: $26 + 30 \approx 55.8$. Electrons contribute essentially nothing to mass. Mass lives in the nucleus; *chemistry* (bonding, reactions) lives in the electrons.

2.4 The periodic table is a map, not a list

The periodic table arranges elements left-to-right by atomic number. Its genius is the **columns**: elements in a column behave alike because they have the same number of outer electrons. Column 1: metals forming $+1$ ions (Na). Column 2: metals forming $+2$ ions (Mg, Ca). Column 16: nonmetals forming -2 ions (O). You do **not** memorize the table. You memorize eight residents.

2.5 The Lunar Eight

These eight elements make up about 99% of the Moon's crust.

Element	Symbol	Z	Mass (g/mol)	Type
Oxygen	O	8	16.0	nonmetal
Sodium	Na	11	23.0	metal
Magnesium	Mg	12	24.3	metal
Aluminum	Al	13	27.0	metal
Silicon	Si	14	28.1	metalloid
Calcium	Ca	20	40.1	metal
Titanium	Ti	22	47.9	metal
Iron	Fe	26	55.8	metal

(Na is Latin *natrium*; Fe is *ferrum*—that’s why those symbols don’t match the English names.)

2.6 The Moon is an oxygen warehouse

By weight, lunar soil is ~42% oxygen. Although oxygen atoms are light, oxygen ions (O^{2-}) are relatively large and form the framework of most lunar minerals. The smaller metal ions, such as Mg^{2+} , Fe^{2+} , Ca^{2+} , and Al^{3+} , occupy organized sites within this framework. Because oxygen ions dominate the crystal structure, they account for roughly 90% of the volume of many lunar mineral lattices.

Think of a lunar mineral as a three-dimensional scaffold built mostly from oxygen atoms. Metal ions such as silicon, magnesium, iron, calcium, and aluminum occupy organized sites within this scaffold, creating the distinct crystal structures that we recognize as lunar minerals.

Exercises:

1. An atom has 22 protons, 26 neutrons, 22 electrons. (a) What element? (b) Approximate mass? (c) Neutral?
2. Explain how lunar soil can be 42% oxygen by weight but ~90% by volume, using *mass* and *size*.
3. A 1 kg lunar sample is 20% Si and 7% Al by weight. Grams of each?

3 Electrons, Ions, and the Logic of Charge

3.1 Electron shells: seating around the nucleus

Electrons fill **shells**: concentric seating tiers from the inside out: shell 1 holds up to **2** electrons, shell 2 up to **8**, shell 3 up to **8** (for our purposes; the full truth is messier and unnecessary here).

Element	Z	Shell 1	Shell 2	Shell 3	Valence e ⁻
O	8	2	6	—	6
Na	11	2	8	1	1
Mg	12	2	8	2	2
Al	13	2	8	3	3
Si	14	2	8	4	4

The outermost electrons are the **valence electrons**: the atom's hands. They do all the chemistry. The periodic table's columns now make sense: same column = same valence count = same behavior.

3.2 The octet rule: atoms want full shells

Atoms are most stable with a **full outer shell** (8 electrons, or 2 for tiny shell 1): the **octet rule**. **Metals** (1–3 valence electrons) most cheaply **lose** them, becoming **cations** (+).

Sodium: 2, 8, 1 → drop one → 2, 8 full, but 11 protons vs. 10 electrons → net +1 → Na⁺.

Nonmetals (6–7 valence electrons) **gain**, becoming **anions** (–).

Oxygen: 2, 6 → grab two → 2, 8 full; 8 protons vs. 10 electrons → O^{2–}.

3.3 Ion size: the second rule of minerals

When an atom loses electrons it shrinks; when it gains, it swells. Approximate ionic radii in picometers:

Ion	Radius (pm)	Size class
O ^{2–}	140	huge (why rocks are mostly oxygen by volume)
Na ⁺	102	large
Ca ²⁺	100	large
Fe ²⁺	78	medium
Mg ²⁺	72	medium
Ti ⁴⁺	61	small–medium
Al ³⁺	54	small
Si ⁴⁺	40	tiny

(1) Fe²⁺ ≈ Mg²⁺ in size, same charge → they will swap freely in crystals.

(2) $\text{Ca}^{2+} \approx \text{Na}^+$ in size but different charge $\rightarrow$ swapping needs an accounting trick.

(3) Si^{4+} is tiny $\rightarrow$ it fits the smallest gaps between oxygens.

Exercises:

1. Using shell counts, show why Mg forms +2 and Al forms +3.
2. Predict neutral formulas: (a) Ca + O, (b) Na + O, (c) Ti + O, (d) Fe^{2+} + O, (e) Si + O.
3. Compute the electron count in O^{2-} , Ca^{2+} , Fe^{2+} , Si^{4+} .
4. Which two lunar cations are nearly interchangeable in size *and* charge? Which pair matches in size but not charge?
5. Explain in energy terms why Na^+ exists but Na^{2+} never appears in minerals.

4 Bonds, Formulas, and Charge Balance

4.1 Two kinds of glue

Ionic bonding. Opposite charges attract. A cation and an anion side by side are bonded by pure electrostatics. Most metal-oxygen contacts in minerals (Ca-O, Mg-O, Fe-O, Na-O) are essentially ionic. Ionic bonds are strong but *undirectional*: the ion attracts neighbors equally in all directions, which is why ionic solids pack into neat geometric lattices.

Covalent bonding. Two atoms *share* an electron pair, each counting it toward its octet. Sharing creates a strong, *directional* bond.

The Si-O bond: roughly half ionic, half covalent, combining electrostatic strength with covalent directionality—the strongest, most stubborn bond in rock chemistry. Consequences: silicates dominate every rocky crust. Silicate minerals are usually hard and resistant to melting because their structures are built from strong silicon–oxygen bonds. However, their hardness and melting behavior depend on how the Si–O units are arranged in the crystal.

4.2 How to read a formula

CaAl₂Si₂O₈ decodes as: Ca = one calcium (no subscript means 1); Al₂ = two aluminums; Si₂ = two silicons; O₈ = eight oxygens. Total: 13 atoms per **formula unit**, the smallest whole recipe. A real crystal repeats it $\sim 10^{20}$ times.

Parentheses with commas, as in (Mg, Fe)₂SiO₄, mean “this position holds Mg *or* Fe, in any proportion, totaling 2 atoms”.

4.3 The charge-balance

Law: Every mineral formula is electrically neutral. Total positive charge = total negative charge, exactly.

Procedure: (1) list each ion, its count, its charge; (2) multiply count $\times$ charge; (3) sum. Must be **zero**.

Auditing plagioclase, CaAl₂Si₂O₈

Ion	Count	Charge	Subtotal
Ca ²⁺	1	+2	+2
Al ³⁺	2	+3	+6
Si ⁴⁺	2	+4	+8
O ²⁻	8	−2	−16
		Sum	+16 − 16 = 0 ✓

Plagioclase is a legal mineral.

4.4 Formula mass

The **formula mass** (molar mass) is the sum of atomic masses in one formula unit, in g/mol.

Formula mass and oxygen fraction of ilmenite FeTiO_3 :

$\text{Fe} = 55.8$; $\text{Ti} = 47.9$; $\text{O} = 3 \times 16.0 = 48.0$. **Total: 151.7 g/mol.**

Oxygen fraction: $48.0 \div 151.7 = 0.316 = \mathbf{31.6\% \text{ by weight.}}$

All four minerals (Mg endmembers for the solid solutions):

Mineral	Formula	Formula mass	% oxygen
Plagioclase	$\text{CaAl}_2\text{Si}_2\text{O}_8$	278.3	46.0
Forsterite	Mg_2SiO_4	140.7	45.5
Ilmenite	FeTiO_3	151.7	31.6

Exercises:

1. Perform full written audits (table format) of Mg_2SiO_4 , FeTiO_3 , $\text{CaAl}_2\text{Si}_2\text{O}_8$.
2. Compute formula masses of albite $\text{NaAlSi}_3\text{O}_8$ and fayalite Fe_2SiO_4 , showing every multiplication.
3. Compute % iron by mass in fayalite.
4. Spinel MgAl_2O_4 occurs in lunar rocks. Audit it, then compute its formula mass.
5. Challenge: using only charge balance, predict the formula of a mineral with one Ca^{2+} , one Ti^{4+} , and O^{2-} . (It exists: perovskite.)

5 The Silicate Tetrahedron

5.1 Why silicon and oxygen rule the crust

Silicon and oxygen are the two most abundant elements in every rocky crust, and their bond is the strongest in geology. Nature's inevitable move: build almost everything from silicon-oxygen units. Three of our four minerals are silicates.

5.2 Meet the tetrahedron

Take one tiny Si^{4+} (40 pm) and pack four huge O^{2-} (140 pm) around it as tightly as geometry allows: four spheres around a tiny center form a **tetrahedron** (a pyramid with triangular base). The silicon disappears into the central cavity. **Charge:** $4(-2) + (+4) = -4$, so the unit is SiO_4^{4-} . It is a negatively charged brick that must be balanced by nearby cations. Every silicate mineral is, at heart, *negatively charged Si-O bricks glued together by positive metal ions*.

5.3 Corner-sharing: the master variable

Two tetrahedra can **share a corner oxygen**—one oxygen belonging to both silicons at once. Every shared oxygen means fewer total oxygens per silicon. Count it (each shared O counts $\frac{1}{2}$ to each Si):

- **No sharing:** each Si owns 4 oxygens $\rightarrow \text{O:Si} = 4.0$
- **Two corners shared:** $2 \text{ whole} + 2 \times \frac{1}{2} = 3 \rightarrow \text{O:Si} = 3.0$
- **All four shared:** $4 \times \frac{1}{2} = 2 \rightarrow \text{O:Si} = 2.0$

Structure	Sharing	O:Si	Mineral	Family name
Isolated islands	0 corners	4.0	Olivine $(\text{Mg, Fe})_2\text{SiO}_4$	nesosilicate
Single chains	2 corners	3.0	Pyroxene $(\text{Ca, Mg, Fe})\text{SiO}_3$	inosilicate
3D framework	4 corners	2.0	Plagioclase $\text{CaAl}_2\text{Si}_2\text{O}_8$	tectosilicate

5.4 What each architecture implies

Olivine-islands. Isolated SiO_4^{4-} bricks glued *entirely* by $\text{Mg}^{2+}/\text{Fe}^{2+}$. Per brick we need +4 of glue: exactly two 2+ cations $\rightarrow (\text{Mg, Fe})_2\text{SiO}_4$. No continuous Si-O network in any direction $\rightarrow$ no flat breakage planes (fractures like glass), and it crystallizes at the highest temperatures.

Pyroxene-chains. Tetrahedra link into infinite single chains, like rebar through the crystal. Chains are internally mighty but held to *each other* only by weaker metal-oxygen glue. Per SiO_3 unit the charge is $4 - 6 = -2$, needing one 2+ cation $\rightarrow (\text{Mg, Fe, Ca})\text{SiO}_3$. The crystal breaks *between* chains: two clean cleavage planes near 90° .

Plagioclase-framework. Every corner shared, a complete 3D network, strong in all directions.

5.5 The odd one out: ilmenite the oxide

Ilmenite FeTiO_3 contains **no silicon**. It is an **oxide**: sheets of oxygen with alternating layers of Fe^{2+} and Ti^{4+} between. No Si-O super-bonds anywhere; its strongest glue is far weaker Fe-O and Ti-O attraction. Ilmenite's oxygen is guarded by weaker bonds.

Exercises:

1. Draw a tetrahedron (or describe one precisely) and verify its -4 charge.
2. Two tetrahedra share one corner. Count atoms.
3. Prove chain silicates have $\text{O}:\text{Si} = 3.0$.
4. Per SiO_3 chain unit (charge -2), which single lunar cations could serve as glue? List all that work.
5. Compute $\text{O}:(\text{Si}+\text{Al})$ for albite $\text{NaAlSi}_3\text{O}_8$. Framework, chain, or island?
6. Ilmenite has no Si. State its structural family and its strongest bond type.

6 Crystals: Architecture Determines Everything

6.1 What a crystal actually is

A **crystal** is matter whose atoms repeat in a perfect 3D pattern. The smallest repeating box is the **unit cell**; stack it like bricks $\sim 10^{20}$ times and you have a visible grain. Everything observable about a mineral, shape, cleavage, color, is the unit cell's architecture amplified a quintillion times.

6.2 The parking-map view of minerals

Between packed oxygen spheres are gaps of three sizes: **tetrahedral holes** (tiny, between 4 oxygens: fit Si^{4+} , Al^{3+}); **octahedral holes** (medium, between 6: fit Mg^{2+} , Fe^{2+} , Ti^{4+}); **large cavities** (roomy: fit Ca^{2+} , Na^{+}).

A **mineral formula is a parking map**: which ions occupy which holes in the oxygen packing. Two ions can trade spots only if size AND charge bookkeeping allow.

6.3 Cleavage: crystals break where bonds are weak

Pyroxene: unbreakable chains, weakly glued side-by-side $\rightarrow$ splits between chains $\rightarrow$ two flat cleavages at $\sim 87^\circ/93^\circ$.

Olivine: isolated islands, glue similar in all directions $\rightarrow$ no preferred plane $\rightarrow$ **conchoidal fracture** (curved, glassy).

Plagioclase: strong framework with slightly weaker sheets.

Ilmenite: breaks along metallic surfaces.

6.4 Density: the property that will shape the Moon

Density = mass $\div$ volume; the levers are ion mass and packing tightness.

Material	Density (g/cm^3)	Why
Plagioclase	2.73	open framework, no iron
<i>Lunar magma (liquid)</i>	~ 2.9	<i>remember this number!</i>
Forsterite (Mg-olivine)	3.22	dense packing, medium ions
Pyroxene	3.2–3.5	dense chains + some Fe
Fayalite (Fe-olivine)	4.39	Fe (55.8) replaces Mg (24.3)
Ilmenite	4.72	heavy Fe + Ti, tight oxide packing

Stare at the second row. Plagioclase is the *only* mineral in this table lighter than lunar magma. Plagioclase in a magma ocean **floats**; everything else **sinks**.

Why Fe-olivine is denser than Mg-olivine Forsterite Mg_2SiO_4 and fayalite Fe_2SiO_4 have *identical* structures. Formula masses: 140.7 vs. $2(55.8) + 28.1 + 64.0 = 203.7$. Same box, 45% more mass $\rightarrow \sim 37\%$ denser (4.39 vs. 3.22; the box also shrinks slightly). **Composition tunes density without touching architecture**—measuring a lunar olivine's density tells you its Fo#. Properties are chemistry made visible.

6.5 Color: why the Moon has a face

Fe^{2+} and Ti absorb visible light; minerals rich in them are dark, minerals without them pale.

Plagioclase: virtually Fe-free $\rightarrow$ white/pale, translucent $\rightarrow$ the bright highlands.

Pyroxene: Fe-rich $\rightarrow$ dark brown-black $\rightarrow$ the dark maria.

Olivine: some Fe $\rightarrow$ bottle green (gem: peridot).

Ilmenite: Fe *and* Ti $\rightarrow$ opaque metallic black $\rightarrow$ the darkest maria. Look at the Moon tonight: the naked-eye brightness pattern is a map of iron chemistry, drawn at planetary scale, unchanged for billions of years.

Exercises:

1. Assign parking spots: Si^{4+} , Al^{3+} , Mg^{2+} , Fe^{2+} , Ti^{4+} , Ca^{2+} , Na^{+} $\rightarrow$ tetrahedral / octahedral / large cavity.
2. Predict the breakage of pyroxene vs. olivine; justify via architecture.
3. Fayalite vs. forsterite: densities 4.39 vs. 3.22. Explain in two sentences with numbers.
4. Order the four minerals plus lunar magma (2.9) by density; mark the one item that floats.
5. The maria look dark from your backyard. Trace the observation down through rock type $\rightarrow$ dominant minerals $\rightarrow$ the responsible ion.
6. A pale translucent lunar grain shows two $\sim 90^\circ$ cleavages; a dark grain shows the same. Identify each.

7 Moles and Stoichiometry: Counting Atoms by Weighing Rocks

7.1 The counting problem

A gram of rock holds $\sim 10^{22}$ atoms, too many to count singly. Chemists count the way bankers count coins: **by weighing**. A bank doesn't count 10,000 pennies; it weighs 25 kg of pennies and *knows*. Chemistry's standard coin roll is the **mole**.

7.2 The mole

1 mole = 6.022×10^{23} **particles** (Avogadro's number), and **one mole of anything weighs its formula mass in grams**:

1 mol O = 16.0 g;

1 mol Fe = 55.8 g;

1 mol ilmenite = 151.7 g.

The bridge equation, most used:

$$\text{moles} = \text{grams} \div \text{formula mass (g/mol)}$$

Weigh a rock $\rightarrow$ divide $\rightarrow$ you have *counted its atoms*.

7.3 Subscripts are mole ratios

One formula unit of FeTiO_3 has 3 oxygens per iron; therefore one **mole** of FeTiO_3 has 3 **moles** of O per mole of Fe. Subscripts scale perfectly from atoms to moles, warehouse arithmetic on invisible particles. That is **stoichiometry** (Greek: "element measuring").

The oxygen locked in a kilogram of ilmenite:

How many grams of oxygen are inside 1 kg of pure ilmenite?

1. grams $\rightarrow$ moles of mineral: $1000 \div 151.7 = \mathbf{6.59}$ mol FeTiO_3
2. moles of mineral $\rightarrow$ moles of O (subscript $\times 3$): $6.59 \times 3 = \mathbf{19.78}$ mol O
3. moles $\rightarrow$ grams: $19.78 \times 16.0 = \mathbf{316}$ g of oxygen

Cross-check: 31.6% of 1000 g = 316 g $\checkmark$.

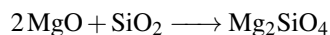
Two independent roads, one answer—stoichiometry is self-auditing. Memorize this **three-step rhythm** (grams $\rightarrow$ moles $\rightarrow$ moles $\rightarrow$ grams).

7.4 Balancing equations

Atoms are never created or destroyed, only rearranged: both sides of an equation must carry equal atoms of each element. Balance by adjusting *coefficients* in front of formulas, never the subscripts inside them.

Building forsterite from oxides

Unbalanced: $\text{MgO} + \text{SiO}_2 \longrightarrow \text{Mg}_2\text{SiO}_4$. Right side has 2 Mg; left has 1 $\rightarrow$ coefficient 2:



Check O: $2 + 2 = 4$ vs. 4 ✓; Mg $2 = 2$ ✓; Si $1 = 1$ ✓.

Similarly $\text{FeO} + \text{TiO}_2 \longrightarrow \text{FeTiO}_3$ (already balanced), and $\text{CaO} + \text{Al}_2\text{O}_3 + 2\text{SiO}_2 \longrightarrow \text{CaAl}_2\text{Si}_2\text{O}_8$.

7.5 Reading a real lunar analysis: oxide weight-percent

Apollo papers report rocks like this (Apollo 17 high-Ti basalt 70017, simplified): SiO_2 38.6, TiO_2 13.0, Al_2O_3 8.7, FeO 18.5, MgO 9.9, CaO 10.4, Na_2O 0.4 (wt%). This does **not** mean piles of loose “FeO” in the rock. It’s an accounting convention: each element expressed *as if* it were its simple oxide.

Read it with trained eyes: iron reported as **FeO**, not **Fe₂O₃**, a lunar signature. And **Na₂O** = 0.4%: the Moon’s missing sodium.

From oxide wt% to mineral content

Basalt 70017 has 13.0 wt% TiO_2 , essentially all in ilmenite. Estimate the ilmenite content.

1. In 100 g rock: $13.0 \text{ g TiO}_2 \rightarrow 13.0 \div 79.9 = 0.163 \text{ mol TiO}_2$
2. Each FeTiO_3 carries one Ti $\rightarrow 0.163 \text{ mol ilmenite}$
3. $0.163 \times 151.7 = \mathbf{24.7 \text{ g}} \rightarrow \mathbf{\text{the rock is } \sim 25\% \text{ ilmenite.}}$

You can now read mining potential straight out of a geochemistry table.

Exercises:

1. Moles, and individual atoms, of oxygen in 500 g of forsterite? (Three-step rhythm; show each step.)
2. Balance: (a) $\text{CaO} + \text{Al}_2\text{O}_3 + \text{SiO}_2 \longrightarrow \text{CaAl}_2\text{Si}_2\text{O}_8$ (b) $\text{MgO} + \text{FeO} + \text{SiO}_2 \longrightarrow \text{MgFeSiO}_4$.
3. Convert 18.5 wt% FeO into grams of iron per 100 g of rock. ($\text{FeO} = 71.8 \text{ g/mol}$.)
4. A low-Ti basalt has 2.5 wt% TiO_2 . Redo the ilmenite estimate; compare mining prospects in one sentence.
5. From 1 kg of pure anorthite, how many grams of aluminum could a perfect refinery extract?
6. Compute oxygen in 1 kg of forsterite two ways (three-step, and %-oxygen shortcut). Confirm agreement.

8 Solid Solutions: What the Parentheses Mean

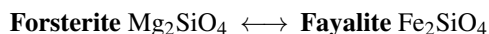
8.1 Minerals are families, not fixed compounds

Most rock-forming minerals are **solid solutions**: continuous families in which certain ions substitute freely for one another inside an unchanging architecture: a parking garage that doesn't care whether a spot holds a gray car or a green car, as long as it's car-sized.

The substitution rules: swapping is free when the ions have (1) similar size, within ~15%; and (2) the same charge. Similar size with *different* charge is allowed only with a compensating second swap.

8.2 Simple substitution: olivine's Mg ↔ Fe slider

Check Mg^{2+} (72 pm) vs. Fe^{2+} (78 pm): same charge ✓, sizes within 8% ✓. The olivine structure genuinely cannot tell them apart, so olivine is a continuous series between two pure **endmembers**:



Composition is reported as the forsterite percentage:

$$\text{Fo\#} = \frac{\text{Mg}}{\text{Mg} + \text{Fe}} \times 100$$

“Fo₇₂” means 72% of the octahedral parking spots hold Mg. **This finally is what (Mg, Fe)₂SiO₄ means:** the parentheses are a slider between endmembers, and the subscript 2 counts the sliding spots per formula unit.

Computing Fo#:

A microprobe measures 1.44 Mg and 0.56 Fe atoms per formula unit.

$$\text{Fo\#} = \frac{1.44}{1.44 + 0.56} \times 100 = \frac{1.44}{2.00} \times 100 = \text{Fo}_{72}$$

Sanity check: $\text{Mg} + \text{Fe} = 2.00$ exactly, matching the subscript in $(\text{Mg}, \text{Fe})_2$. If your atoms don't sum to the subscript, re-measure.

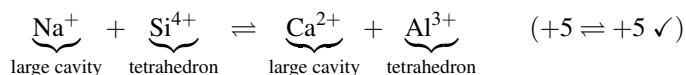
From Fo# back to a formula, then to mass:

Write Fo₆₀ olivine and find its % iron by mass. Fo₆₀: 60% of 2 spots are Mg → $\text{Mg}_1 \cdot 2\text{Fe}_0 \cdot 8\text{SiO}_4$ (decimal subscripts are normal for solid solutions—averages over 10²⁰ unit cells).

Formula mass: $1.2(24.3) + 0.8(55.8) + 28.1 + 64.0 = 29.2 + 44.6 + 28.1 + 64.0 = 165.9$. %Fe = $44.6/165.9 = \mathbf{26.9\%}$.

8.3 Coupled substitution: plagioclase's elegant trick

Na^+ (102 pm) vs. Ca^{2+} (100 pm): size ✓, charge × (+1 vs. +2). A straight swap wrecks charge balance. Plagioclase's solution is a **paired trade executed simultaneously in two parking zones**:



Swap big Na for big Ca (+1 gained in the cavity) *and* tetrahedral Si for Al (−1 in the framework): net zero. Endmembers:



Watch the formulas transform under the trade: albite's (1 Na, 1 Al, 3 Si) becomes anorthite's (1 Ca, 2 Al, 2 Si): one Na→Ca, one Si→Al, oxygens untouched. Audit both endmembers; both balance. The formula pair *encodes* the substitution rule.

Pyroxene plays the same game in two dimensions (Ca-Mg-Fe sharing sites), mapped on the **pyroxene quadrilateral**. Lunar pyroxenes to know by name: **augite** (Ca-rich) and **pigeonite** (Ca-poor).

8.4 The payoff: one number that records the Moon's birth

Earth's plagioclase spans the whole range: beach sands around An₂₀₋₄₀, volcanic rocks higher. Lunar plagioclase crowds the extreme: **An₉₄₋₉₈, nearly pure anorthite**.

Why? Sodium is a **volatile** element: it boils out of rock vapor at temperatures where calcium doesn't flinch. The Moon condensed from incandescent debris splashed out when a Mars-sized body struck the young Earth. In that furnace, sodium largely boiled away; calcium stayed. The Moon was born sodium-starved, so its plagioclase *had* to be calcium-flavored.

Read that as a method, not a fact: **a single measured number, An₉₆, is a fossil of an event 4.5 billion years ago**. Composition is history. This is why geochemists measure obsessively: every solid solution is a recording device.

Exercises:

1. Compute Fo# for (Mg, Fe) pairs: (1.80, 0.20), (1.00, 1.00), (0.30, 1.70). Label each Mg-rich, even, or Fe-rich.
2. Verify by written audit that the coupled substitution conserves charge, and show it converts albite's formula into anorthite's, subscript by subscript.
3. Genesis Rock plagioclase: Ca 0.96, Na 0.04. Idaho granite: Ca 0.25, Na 0.75. Compute both An#.
4. In two sentences: what does the difference say about *sodium* and about *the Moon's origin*?
5. Write the formula of Fo₉₀ olivine; compute its formula mass and % oxygen.
6. Ti⁴⁺ (61 pm) vs. Si⁴⁺ (40 pm): same charge. Does Ti freely enter silicate tetrahedra? Cite the rule it passes or fails, and state what your answer implies about where the Moon's Ti must end up. (It explains ilmenite's existence.)

9 Redox: The Moon's Missing Rust

9.1 Electron bookkeeping

Oxidation = losing electrons. Reduction = gaining electrons. (Mnemonic: **OIL RIG**: Oxidation Is Loss, Reduction Is Gain.) The two always occur together: if something loses electrons, something else catches them. An atom's **oxidation state** is its running electron ledger. The new idea: *some elements have several available states*, and which state you find is **evidence about the environment**.

9.2 Iron's three costumes

State	Name	Electrons	Found in
Fe ⁰	metallic iron	all kept	meteorites, steel, lunar soil grains
Fe ²⁺	ferrous	lost 2	every lunar mineral ; Earth's deep rocks
Fe ³⁺	ferric	lost 3	rust, red deserts— absent on the Moon

Which costume iron wears depends on how aggressively the surroundings steal electrons. Free O₂ and water are relentless electron thieves; environments rich in them are **oxidizing**, environments without them **reducing**.

Earth's surface: O₂ + water everywhere → iron mugged to Fe³⁺ → the planet literally rusts (why Mars and Australian deserts are red).

The Moon: no atmosphere, no water → Fe²⁺ everywhere, and micrometeorite impacts even smelt soil iron to **metallic Fe⁰** droplets nanometers across ("nanophase iron"), which gradually darken exposed soil, so soil *maturity* can be read from its darkness ("space weathering").

9.3 The audit meets redox

Lunar ilmenite passes:

FeTiO₃ with lunar charges: (+2) + (+4) + 3(−2) = 0 ✓. No Fe³⁺ required. Natively lunar-compatible.

Rust fails the lunar audit:

Hematite Fe₂O₃ with lunar Fe²⁺: 2(+2) + 3(−2) = −2 ✗. Hematite *cannot exist* with Fe²⁺; it demands Fe³⁺: 2(+3) + 3(−2) = 0 ✓. Conclusion: **hematite is chemically impossible on the Moon**.

Mixed-state magnetite (solve-through):

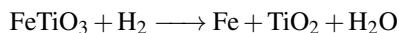
Magnetite Fe₃O₄ mixes Fe²⁺ and Fe³⁺. Let x = number of Fe³⁺:

$$x(+3) + (3-x)(+2) + 4(-2) = 0 \Rightarrow 3x + 6 - 2x - 8 = 0 \Rightarrow x = 2$$

Magnetite is exactly (Fe³⁺)₂(Fe²⁺)O₄. Algebra just did mineralogy, and since it needs Fe³⁺, magnetite in a claimed "lunar" sample is a contamination alarm.

9.4 Redox in reactions: reading the hand-off

Track electrons through reaction:



Iron: $+2 \rightarrow 0$, gained 2 electrons, **reduced** (the point of the process).

Hydrogen: $0 \rightarrow +1$, **oxidized** (the sacrificial donor).

Titanium: $+4 \rightarrow +4$, untouched—its grip on oxygen is too strong for hydrogen.

Oxygen: -2 throughout; it merely changes partners ($\text{Fe} \rightarrow \text{H}$).

You can now *read* a redox reaction like a transaction record: who paid electrons, who received, who sat out.

Exercises:

1. Assign oxidation states to every atom in all four target minerals.
2. Show by audit that Fe_2O_3 requires Fe^{3+} and FeO permits Fe^{2+} .
3. Redo the magnetite algebra from scratch; explain the contamination-alarm logic in one sentence.
4. In $2\text{H}_2\text{O} \longrightarrow 2\text{H}_2 + \text{O}_2$ (electrolysis), assign states to H and O on both sides. What is oxidized? Reduced?
5. Lunar soil darkens with surface exposure. Explain via nanophase Fe^0 .
6. An Antarctic meteorite contains An_{97} plagioclase, Fe^{2+} -only pyroxene, traces of Fe^0 , and zero hydrated or Fe^{3+} minerals. Build the case for lunar origin. (Such meteorites are real.)
7. Apollo 16 soils contain rare grains with Fe^{3+} signatures, interpreted as meteorite debris. Why is “delivered from outside” more plausible than “formed on the Moon”?

10 The Magma Ocean: Why the Moon Has a Face

10.1 The setup: a molten Moon

4.5 billion years ago the Moon assembled hot, hot enough that its outer several hundred kilometers were an ocean of magma: a liquid solution of all our ions at once (O^{2-} , Si^{4+} , Mg^{2+} , Fe^{2+} , Ca^{2+} , Al^{3+} , Ti^{4+} , trace Na^+), density $\approx 2.9 \text{ g/cm}^3$, temperature $\gtrsim 1400^\circ\text{C}$. Then it began to cool. What happens when this soup freezes is the founding story of lunar science.

10.2 Freezing is sequential, and each mineral edits the melt

A mixed liquid doesn't freeze all at once (seawater doesn't freeze like pure water: ice forms and leaves brine behind). Minerals crystallize in order of formation temperature, the simplified **Bowen sequence** for our cast:

1. **Olivine** first ($\gtrsim 1400^\circ\text{C}$): small isolated tetrahedra + abundant early Mg, easiest to assemble from hot soup
2. **Pyroxene** next ($\sim 1300^\circ\text{C}$): chains take more organizing
3. **Plagioclase (anorthite)** later ($\sim 1200^\circ\text{C}$): frameworks assemble last among the majors
4. **Ilmenite** near the end: only from late, cooled, enriched liquid

Crucially, each crystal **subtracts its ingredients from the remaining melt**. Early olivine strips Mg and Fe; the leftover liquid grows relatively richer in Ca and Al, and, because ilmenite hasn't formed yet, **Ti keeps concentrating in the shrinking residual melt**. This running edit is **fractional crystallization**, the most important process in planetary geology. (The reason Ti waits: Ti^{4+} is too big for tetrahedra, so no major early mineral takes it. It has nowhere to go but the leftovers until enough accumulates to build its own mineral, ilmenite.)

10.3 Density is destiny

Every new crystal immediately faces: Magma: 2.9.

- **Olivine (3.2–4.4): sinks.** Rains down to build the deep **mantle**.
- **Pyroxene (~ 3.4): sinks.** Builds the upper mantle.
- **Anorthite (2.73): floats:** the only buoyant crystal in the story. Anorthite crystals rose like froth and accreted into a globe-wrapping raft of nearly pure plagioclase rock: **anorthosite** (rock, not mineral!). That raft, ~ 4.4 billion years old, is today's **bright highlands crust**. When you look at the Moon's bright regions, you are seeing *flotation scum on a frozen ocean*, preserved four billion years.
- **Ilmenite (4.72): sinks**, late, forming dense Ti-rich layers in the upper mantle.
- The last dregs ($\sim 1\%$), enriched in everything no mineral wanted, froze as a layer called **KREEP** (K, Rare Earth Elements, P).

10.4 Act two: the maria (the part everyone gets backwards)

The magma ocean built highlands + mantle by ~ 4.4 Gyr ago. The dark **maria** came *later*, by a different mechanism:

1. Giant impacts excavated enormous basins in the anorthosite crust (~ 4.0 – 3.8 Gyr).

2. Radioactive heating re-warmed the interior, **partially re-melting** the sunken olivine–pyroxene mantle ($\sim 3.8\text{--}3.1$ Gyr).
3. That re-melt, **basaltic magma**, leaked up through basin-floor fractures and flooded them, freezing into thin dark plains of **mare basalt**: mostly pyroxene + plagioclase, with olivine, and abundant ilmenite *only where* the re-melting sampled those sunken Ti-rich layers.

Hence **high-Ti basalts** (Apollo 11, 17, up to $\sim 13\%$ TiO_2 , $\sim 25\%$ ilmenite) versus **low-Ti basalts** (Apollo 12, 15). Titanium is patchy because the buried ilmenite layers are patchy.

The timeline trap (test yourself):

Olivine and pyroxene crystallized FIRST (4.4+ Gyr), yet mare basalts made of them are the YOUNGEST major surfaces (3.8–3.1 Gyr). Contradiction? No: the maria are *re-melts*. The minerals crystallized twice, once in the ocean (then sank), then again from second-generation magma a half-billion years later. First-formed minerals, youngest rocks.

10.5 The grand synthesis

Why are the highlands bright?

Anorthite is Fe-poor $\rightarrow$ pale $\rightarrow$ because it floated $\rightarrow$ as nearly pure An_{96} $\rightarrow$ because Na is volatile and was baked away at the Moon's birth $\rightarrow$ crystallizing from a magma ocean.

Why are the maria dark?

Fe^{2+} -rich pyroxene + ilmenite $\rightarrow$ from re-melted sunken mantle $\rightarrow$ iron kept as Fe^{2+} because the Moon is reducing.

Why is titanium patchy?

Ti^{4+} fits no early mineral $\rightarrow$ concentrated by fractional crystallization $\rightarrow$ froze into late ilmenite layers $\rightarrow$ tapped by only some mare re-melts.

Exercises:

1. Write the four minerals in crystallization order, then density order. Circle the one that is both *late* and *light*, and state why that combination, uniquely, builds a crust.
2. If anorthite's density were 3.1 g/cm^3 , narrate tonight's Moon: crust of what? Highlands where? Would "anorthosite" exist as a word?
3. Genesis Rock (15415): nearly pure An_{96} anorthosite, 4.4 Gyr old, found in the highlands. Show each descriptor is a *prediction* of the magma-ocean model.
4. Resolve the timeline trap in ≤ 3 sentences without the word "remelt", force yourself to explain the mechanism.
5. Why is ilmenite abundant in some maria and nearly absent in others? Chain the full argument: ion size $\rightarrow$ fractional crystallization $\rightarrow$ layers $\rightarrow$ partial re-melting.
6. The far side has thicker anorthosite crust and few maria. Predict its average surface Ti content and defend it.

A The Booklet on One Page

Unit	The one idea	The skill
0	Matter is countable	scientific notation, %, ratios
1	Eight elements build the Moon	the Lunar Eight table
2	Atoms trade electrons predictably	the 8 ion charges; size classes
3	Formulas sum to zero charge	the charge-balance audit; formula mass
4	One brick, shared corners	O:(Si+Al) structural fingerprint
5	Architecture → properties	density / cleavage / color reasoning
6	Count atoms by weighing	grams $\rightleftharpoons$ moles $\rightleftharpoons$ moles $\rightleftharpoons$ grams
7	Minerals are sliding families	Fo#, An#, substitution rules
8	Oxidation state is evidence	redox audits; OIL RIG
9	The ocean froze in sorted layers	fractional crystallization + density
10	Weak bonds get mined	ISRU mass balance

B Master Data Tables

The Lunar Eight (mass g/mol, ion): O 16.0 (−2) · Na 23.0 (+1) · Mg 24.3 (+2) · Al 27.0 (+3) · Si 28.1 (+4) · Ca 40.1 (+2) · Ti 47.9 (+4) · Fe 55.8 (+2)

Formula masses (g/mol): anorthite 278.3 · albite 262.2 · enstatite 100.4 · forsterite 140.7 · fayalite 203.7 · ilmenite 151.7 · FeO 71.8 · TiO₂ 79.9 · SiO₂ 60.1 · MgO 40.3 · CaO 56.1 · Al₂O₃ 102.0 · H₂O 18.0 · O₂ 32.0

Densities (g/cm³): anorthite 2.73 · lunar magma ~2.9 · forsterite 3.22 · pyroxene 3.2–3.5 · fayalite 4.39 · ilmenite 4.72

Ionic radii (pm): O^{2−} 140 · Na⁺ 102 · Ca²⁺ 100 · Fe²⁺ 78 · Mg²⁺ 72 · Ti⁴⁺ 61 · Al³⁺ 54 · Si⁴⁺ 40